

PDS Search LDD Form

Scientific Discipline

Information about the scientific discipline and research context.

Primary Field What are the different scientific fields of study.

- ☐ **Astrobiology** Study of life in the universe
 - ☐ **Astronomy Observation** Observation of celestial objects
 - ☐ **Astronomy Radar** Radar observations of celestial objects
 - ☐ **Astronomy Exoplanets** Study of exoplanets
 - ☐ **Astronomy Spectroscopy** Spectroscopy of celestial objects
 - ☐ **Geology Geophysics** Geophysics of the Earth and other planets
 - ☐ **Geology Seismology** Seismology of the Earth and other planets
 - ☐ **Geology Surface** Surface studies of the Earth and other planets
 - ☐ **Geology Crater Counting** Crater counting of the Earth and other planets
 - ☐ **Geology Interior Studies** Interior studies of the Earth and other planets
 - ☐ **Dynamics** Dynamics of the Earth and other planets
 - ☐ **Laboratory Geochemistry** Geochemistry of physical samples
 - ☐ **Laboratory Cosmochemistry** Cosmochemistry of physical samples
 - ☐ **Laboratory Age Dating** Age dating of physical samples
 - ☐ **Laboratory Analytic Chemistry** Analytic chemistry of physical samples
 - ☐ **Laboratory Spectral Analysis** Spectral analysis of physical samples
 - ☐ **Atmospheric Study** Atmospheric studies of the Earth and other planets
 - ☐ **Impact** Measuring and categorizing impact craters and rates
 - ☐ **Hydrology** Fluvial mechanics
 - ☐ **Exoplanets** Detection and characterizing exoplanets
 - ☐ **Modeling** Modeling of the Earth and other planets
 - ☐ **Magnetospheres** Magnetospheres of the Earth and other planets
 - ☐ **Space Physics** Space physics of the Earth and other planets
 - ☐ **Other** Any field not covered by other values
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Investigation Technique What method of scientific investigations are used.

- ☐ **Spectroscopy** Collection of spectra
- ☐ **Modeling** Computer-based simulation or modeling
- ☐ **Mapping** Collection of maps
- ☐ **Radar** Collection of radar data
- ☐ **Seismology** Collection of seismic data
- ☐ **Ground based observation** Data that was derived from Earth-based telescopes
- ☐ **Gravimetry** Collection of gravity data
- ☐ **Remote Sensing** Various data collected via remote sensing techniques
- ☐ **Geochemistry** Collection of geochemical/cosmochemical data
- ☐ **Laboratory** Various laboratory work
- ☐ **Artificial Intelligence** AI and machine learning techniques
- ☐ **ICPMS** Collection of inductively coupled plasma mass spectrometry data
- ☐ **Other** Any investigation technique not covered by other values

Study Focus Specific area of study within the scientific discipline.

- ☐ **Surface** Study focused on planetary geology such as features or properties
 - ☐ **Interior** Study focused on internal structure or processes
 - ☐ **Atmosphere** Study focused on atmospheric properties or dynamics
 - ☐ **Magnetosphere** Study focused on magnetic field properties or interactions
 - ☐ **Orbital Dynamics** Study focused on orbital properties or behaviors
 - ☐ **Composition** Study focused on chemical or mineral composition
 - ☐ **Plasma** Study of plasma and ionospheres
 - ☐ **Space Weather** Study of the change in surfaces due to space weather effects
 - ☐ **Planetary Protection** Study focused on planetary protection
 - ☐ **Planetary Rings** Study focused on ring and ring-moon systems
 - ☐ **Planetary Resources** In-situ resource characterization and utilization
 - ☐ **Other** Any focus not covered by other values
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Target Object

Information about the target of observation, whether it is a celestial body, feature, or other object.

Object Name Name of the object

Value: _____

Parent Body Parent body of the object

Value: _____

Scope Scope of the data collection

- ☐ **Global** Data collected from the entire object
 - ☐ **Regional** Data collected from a large region of the object
 - ☐ **Local** Data collected from a small location on the object
 - ☐ **Other** Any scope not covered by other values
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Object Type Select the categories within which the data would fit into

Planet Planet

- ☐ **Gas Giant** Large planets composed mainly of gas
 - ☐ **Terrestrial** Rocky planets similar to Earth
 - ☐ **Interior** Planets closer to the Sun than Earth
 - ☐ **Other** Any planet type not covered by other values
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Particles and Fields Fields of an object

- ☐ **Particles** Particles in the atmosphere
 - ☐ **Cosmic Rays** Cosmic rays in the atmosphere
 - ☐ **X-Rays** X-rays in the atmosphere
 - ☐ **Gamma-Rays** Gamma-rays in the atmosphere
 - ☐ **Electromagnetic Waves** Electromagnetic waves in the atmosphere
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Dust Dust

- ☐ **Meteoroids** Dust from meteoroids
 - ☐ **IPD** Interplanetary Dust
 - ☐ **Comet** Dust from comets
 - ☐ **Other** Other dust
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Solar Component Solar component

- ☐ **Corona** Corona
 - ☐ **Solar Wind** Solar Wind
 - ☐ **Exoplanets** Exoplanets
 - ☐ **Heliosphere** Heliosphere
 - ☐ **Core** Core
 - ☐ **Radiative Zone** Radiative Zone
 - ☐ **Convective Zone** Convective Zone
 - ☐ **Photosphere** Photosphere
 - ☐ **Chromosphere** Chromosphere
 - ☐ **Other** Any solar component not covered by other values
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Exoplanet Type of an exoplanet

- ☐ **Terrestrial** Rocky planets, up to 1.5 Earth radii
 - ☐ **Gas Giants** Larger than 10 Earth radii
 - ☐ **Ice Giants** 4-6 Earth radii
 - ☐ **Super-Earths** 1.5 to 2 Earth radii
 - ☐ **Mini-Neptunes** 2-04 Earth radii
 - ☐ **Lava Worlds** Molten rock
 - ☐ **Ocean Worlds** Global oceans
 - ☐ **Rogue Planets** Planets not bound to a star
 - ☐ **Other** Other categories
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Small Body Small body

Asteroid Asteroid

- ☐ **Near Earth** Asteroids with orbits that bring them close to Earth
 - ☐ **Trojan** Asteroids that share an orbit with a planet
 - ☐ **Active** Asteroids that show comet-like activity
 - ☐ **Main Belt** Asteroids in the main asteroid belt
 - ☐ **PHA** Potential Hazardous Asteroid
 - ☐ **Other** Any asteroid type not covered by other values
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TNO Trans-Neptunian Object

- ☐ **Classical** Objects in the classical Kuiper belt
 - ☐ **Resonant** Objects in orbital resonance with Neptune
 - ☐ **Scattered** Objects scattered by Neptune
 - ☐ **Detached** Objects with orbits not strongly influenced by Neptune
 - ☐ **Other** Any TNO type not covered by other values
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Centaur Centaur

- ☐ **Active** Centaurs showing comet-like activity
 - ☐ **Inactive** Centaurs without comet-like activity
 - ☐ **Other** Any centaur type not covered by other values
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KBO Kuiper Belt Object

- ☐ **Classical** Objects in the classical Kuiper belt
 - ☐ **Resonant** Objects in orbital resonance with Neptune
 - ☐ **Scattered** Objects scattered by Neptune
 - ☐ **Other** Any KBO type not covered by other values
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Oort Cloud Oort Cloud Object

- ☐ **Inner** Objects in the inner Oort cloud
 - ☐ **Outer** Objects in the outer Oort cloud
 - ☐ **Other** Any Oort cloud type not covered by other values
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Comet Comet

- ☐ **Jupiter Family** Comets that orbit Jupiter
 - ☐ **Single Pass** Comets that pass through the inner solar system
 - ☐ **Other** Any comet type not covered by other values
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Satellite Satellite

- ☐ **Icy Moons** Icy moons of a planet
- ☐ **Galilean Moons** Galilean moons of Jupiter
- ☐ **Irregular Moons** Irregular moons of a planet
- ☐ **Captured Moons** Captured moons of a planet
- ☐ **Shepherding Moon** Shepherding moon of a planet
- ☐ **Other** Any satellite type not covered by other values
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Feature Feature

Name Name of the feature

Value:

Type Type of the feature

- ☐ **Crater** Impact depression on a surface
- ☐ **Mountain** Elevated landform
- ☐ **Valley** Elongated depression
- ☐ **Mare** Dark plains on the Moon
- ☐ **Volcano** Volcanic landform
- ☐ **Cloud** Atmospheric cloud formation
- ☐ **Other** Any feature type not covered by other values
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Atmosphere Atmosphere of a celestial body

Layer Layer of the atmosphere

- ☐ **Troposphere** Lowest layer of the atmosphere
- ☐ **Stratosphere** Layer above the troposphere
- ☐ **Ionosphere** Layer containing ions
- ☐ **Exosphere** Outermost layer of the atmosphere
- ☐ **Other** Any atmospheric layer not covered by other values
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Study Type Type of study of the atmosphere

- ☐ **Temperature Profile** Temperature profile of the atmosphere
- ☐ **Scale Height** Scale height of the atmosphere
- ☐ **Density Profile** Density profile of the atmosphere
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Rings Rings of an object

Ring Type Type of ring

- ☐ **D** D ring
 - ☐ **C** C ring
 - ☐ **B** B ring
 - ☐ **A** A ring
 - ☐ **F** F ring
 - ☐ **G** G ring
 - ☐ **E** E ring
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Ring Division Division of the ring

- ☐ **Cassini Division** Cassini Division
 - ☐ **Encke** Encke Division
 - ☐ **Keeler** Keeler Division
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Ring Feature Feature of the ring

- ☐ **Spiral Bending Waves** Spiral Bending Waves
 - ☐ **Spokes** Spokes
 - ☐ **Spiral Density Waves** Spiral Density Waves
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Physical Properties

Information about the physical properties of the object under study.

Material Type Material type of the object

- ☐ **Rock** Solid mineral material
 - ☐ **Ice** Frozen water or other volatiles
 - ☐ **Gas** Gaseous material
 - ☐ **Dust** Fine particulate material
 - ☐ **Regolith** Surface layer of loose material
 - ☐ **Liquid** Liquid state material
 - ☐ **Other** Any material type not covered by other values
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Process Type Type of process being studied.

- ☐ **Impact** Collision processes
 - ☐ **Volcanic** Volcanic activity
 - ☐ **Tectonic** Crustal deformation processes
 - ☐ **Atmospheric** Atmospheric processes
 - ☐ **Erosional** Surface degradation processes
 - ☐ **Other** Any process type not covered by other values
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Composition Primary composition of the material being studied.

- ☐ **Silicate** Composed primarily of silicate minerals
 - ☐ **Metal** Composed primarily of metallic elements
 - ☐ **Organic** Composed primarily of organic compounds
 - ☐ **Icy** Composed primarily of water or other volatile ices
 - ☐ **Gaseous** Composed primarily of gas
 - ☐ **Mixed** Composed of a mixture of different materials
 - ☐ **Other** Any composition not covered by other values
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Environmental Context

Information about the environmental conditions at the study location.

Temperature Regime Temperature conditions at the study location.

- ☐ **Cryogenic** Extremely low temperatures, typically below 120K
 - ☐ **Cold** Below freezing but not extreme
 - ☐ **Temperate** Near-surface temperatures typical of Earth
 - ☐ **Warm** Above freezing and below boiling point
 - ☐ **Hot** Above boiling point
 - ☐ **Other** Any temperature regime not covered by other values
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Pressure Conditions Pressure conditions at the study location.

- ☐ **Vacuum** Essentially no atmospheric pressure
 - ☐ **Very Low** Minimal atmospheric pressure like Mars
 - ☐ **Earth Like** Similar to Earth's atmospheric pressure
 - ☐ **High** Higher than Earth's atmospheric pressure
 - ☐ **Extreme** Very high pressure like Venus surface or gas giant interiors
 - ☐ **Variable** Significant pressure variations or gradients
 - ☐ **Other** Any pressure condition not covered by other values
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Radiation Environment Radiation conditions at the study location.

- ☐ **Low** Minimal radiation exposure like Earth surface
 - ☐ **Moderate** Similar to Earth orbit or Mars surface
 - ☐ **High** Significant radiation like Jupiter radiation belts
 - ☐ **Extreme** Very high radiation levels
 - ☐ **Shielded** Protected or shielded from external radiation
 - ☐ **Other** Any radiation environment not covered by other values
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Atmospheric Presence Presence and nature of atmosphere at the study location.

- ☐ **None** No atmosphere present
 - ☐ **Trace** Very low atmospheric pressure
 - ☐ **Low** Atmospheric pressure similar to Earth's
 - ☐ **High** Atmospheric pressure similar to Earth's
 - ☐ **Extreme** Very high atmospheric pressure like Venus surface
 - ☐ **Variable** Significant atmospheric pressure variations or gradients
 - ☐ **Other** Any atmospheric presence not covered by other values
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Data Collection Information about how the data was collected and its format.

Collection Method Method used to collect the data.

- ☐ **Orbiter** Data collected by an orbiting spacecraft
 - ☐ **Lander** Data collected by a surface lander
 - ☐ **Rover** Data collected by a mobile surface rover
 - ☐ **Telescope** Data collected by Earth-based or space telescopes
 - ☐ **Sample Return** Analysis of returned physical samples
 - ☐ **Laboratory** Data collected in laboratory settings
 - ☐ **Other** Any collection method not covered by other values
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Instrument Type Type of instrument used to collect the data.

- ☐ **Spectrometer** Instrument that measures spectra
 - ☐ **Camera** Imaging device
 - ☐ **Radar** Radio detection and ranging instrument
 - ☐ **Lidar** Light detection and ranging instrument
 - ☐ **Seismometer** Instrument that measures ground motion
 - ☐ **Magnetometer** Instrument that measures magnetic fields
 - ☐ **Other** Any instrument type not covered by other values
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Data Category General category of the collected data.

- ☐ **Altimetry** Elevation data
 - ☐ **Image** Visual representation data
 - ☐ **Photometry** Photometric measurement data
 - ☐ **Map** Spatial representation data
 - ☐ **Model Output** Results from computational models
 - ☐ **Database** Flat rendering of existing data base
 - ☐ **Parameter Data** Numerical data
 - ☐ **Other** Any data category not covered by other values
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Data Format Format of the collected data.

- ☐ **Shape Model DTM** A 3D model of the surface of the object
 - ☐ **Geologic Map** A map of the geologic features of the object
 - ☐ **Thermal Map** A map of the thermal properties of the object
 - ☐ **Mosaic** A composite image of the object
 - ☐ **Spectral Map** A map of the spectral properties of the object
 - ☐ **Gazetteer** A list of feature names
 - ☐ **Time Series** Sequential data collected over time
 - ☐ **Crater** A list of craters on the object
 - ☐ **Light Curve** A list of light curves on the object
 - ☐ **Spectral** A list of spectra on the object
 - ☐ **Physical Properties** A list of physical properties of the object
 - ☐ **Photometric Properties** A list of photometric properties of the object
 - ☐ **Gravity Map** A map of the gravity of the object
 - ☐ **Taxonomies** A list of taxonomies of the object
 - ☐ **Crater Type** A list of irregular crater types on the object
 - ☐ **Other** Any data format not covered by other values
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Temporal Epoch The temporal epoch of the data collection.

Earth Temporal epoch of the Earth

- ☐ **Eoarchean** Earth's 4,031 to 3,600 million years ago
 - ☐ **Paleoarchean** Earth's 3,600 to 3,200 million years ago
 - ☐ **Mesoarchean** Earth's 3,200 to 2,800 million years ago
 - ☐ **Neoarchean** Earth's 2,800 to 2,500 million years ago
 - ☐ **Paleoproterozoic** Earth's 2,500 to 1,600 million years ago
 - ☐ **Mesoproterozoic** Earth's 1,600 to 1,000 million years ago
 - ☐ **Neoproterozoic** Earth's 1,000 to 538.8 million years ago
 - ☐ **Paleozoic** Earth's 238.8 to 251.9 million years ago
 - ☐ **Mesozoic** Earth's 251.9 to 66 million years ago
 - ☐ **Cenozoic** Earth's 66-0 million years ago
 - ☐ **Ohter** Earth's Paleozoic Era
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Moon Temporal epoch of the Moon

- ☐ **Pre-Nectarian** Moon, 4,533 to 3,920 million years ago
 - ☐ **Nectarian** Moon 3,920 to 3,850 million years ago
 - ☐ **Imbrian** Moon 3,850 to 3,200 millino years ago
 - ☐ **Eratosthenian** Moon 3,200 to 1,100 million years ago
 - ☐ **Copernican** Moon 1,100 to 0 million years ago
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Mars Temporal epoch of Mars

- ☐ **Pre-Noachian** Mars 4,500 to 4,100 million years ago
 - ☐ **Noachian** Mars 4,100 to 3,700 million years ago
 - ☐ **Hesperian** Mars 3,700 to 3,000 million years ago
 - ☐ **Amazonian** Mars 3,000 to 0 million years ago
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Vesta Relative timescale for Vesta

- ☐ **Pre-Veneneia** \> 4,100 million years ago
 - ☐ **Veneneia** ~ 4,100 million years ago
 - ☐ **Rheasilvia** ~ 3,900 million years ago
 - ☐ **Marcia** ~3,500 million years ago
-

Ceres Relative timescale for Ceres

- ☐ **Pre-Kerwan** ~4,560 to 4,000 million years ago
 - ☐ **Kerwan** 4,000 to 2,500 million years ago
 - ☐ **Ernutet** ~ 2.500 to 0 million years ago
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Analysis Properties

Information about the analysis methods and properties of the data.

Analysis Method Method used to analyze the data.

- ☐ **Spectroscopy** Collection of spectra
 - ☐ **Modeling** Computer-based simulation or modeling
 - ☐ **Mapping** Collection of maps
 - ☐ **Radar** Collection of radar data
 - ☐ **Seismology** Collection of seismic data
 - ☐ **Gravimetry** Collection of gravity data
 - ☐ **Geochemistry** Collection of geochemical data
 - ☐ **SEM** Collection of scanning electron microscopy data
 - ☐ **TEM** Collection of transmission electron microscopy data
 - ☐ **ICPMS** Collection of inductively coupled plasma mass spectrometry data
 - ☐ **Other** Any analysis method not covered by other values
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Software Used Software used to analyze the data.

- ☐ **Spectroscopy** Collection of spectra
- ☐ **Modeling** Computer-based simulation or modeling
- ☐ **Mapping** Collection of maps
- ☐ **Radar** Collection of radar data
- ☐ **Seismology** Collection of seismic data
- ☐ **Gravimetry** Collection of gravity data
- ☐ **Geochemistry** Collection of geochemical data

- ☐ **SEM** Collection of scanning electron microscopy data
 - ☐ **TEM** Collection of transmission electron microscopy data
 - ☐ **ICPMS** Collection of inductively coupled plasma mass spectrometry data
 - ☐ **Other** Any software not covered by other values
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Processing Steps Processing steps used to analyze the data.

- ☐ **Filtering** Filtering of the data
 - ☐ **Smoothing** Smoothing of the data
 - ☐ **Deconvolution** Deconvolution of the data
 - ☐ **Other** Any processing step not covered by other values
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Uncertainty Measurements Uncertainty measurements of the data.

- ☐ **Standard Deviation** Standard deviation of the data
 - ☐ **Standard Error** Standard error of the data
 - ☐ **Confidence Interval** Confidence interval of the data
 - ☐ **Other** Any uncertainty measurement not covered by other values
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Quality Indicators Quality indicators of the data.

- ☐ **Accuracy** Accuracy of the data
 - ☐ **Precision** Precision of the data
 - ☐ **Other** Any quality indicator not covered by other values
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Research Context

Information about the research context and associated metadata.

Research Goals Research goals of the study

- ☐ **Accuracy** Accuracy of the data
 - ☐ **Precision** Precision of the data
 - ☐ **Other** Any research goal not covered by other values
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Related Publications Related publications of the study

Value:

Keywords Keywords of the study

Value:

Principal Investigator Principal investigator of the study

Value: _____

Institution Institution of the study

Value: _____
